

SENIOR NETWORK INFRASTRUCTURE CONSULTING REPORT

Building a Production-Grade Multi-Vendor
Cisco Network Lab Environment

Leveraging IBM Cloud + Red Hat OpenShift
& Borrowing Best Practices from Palo Alto, Juniper, Fortinet & Others

Prepared for:	i3 Smart Labs Leadership & Engineering Teams
Environment:	IBM Cloud on Red Hat OpenShift
Source Documents:	Vendor Lab Environments Matrix (Dec 2025) + CCIE EI Lab Specs
Date:	September 13, 2026
Classification:	Internal Strategic Advisory

This document provides actionable architecture recommendations, open-source strategies, and an implementation roadmap for building scalable, multi-vendor network labs that support certification, training, PoCs, and production validation.

1. Executive Summary

Your existing Vendor Lab Environments Matrix (December 2025) and the commercial CCIE Enterprise Infrastructure lab offerings (Dell R620/T5610 based EVE-NG + SD-WAN kits priced \$1,000–\$1,150) demonstrate a solid foundation focused primarily on traditional hypervisor-based virtualization (ESXi + EVE-NG / CML). However, with an IBM Cloud + Red Hat OpenShift platform already available, you have a strategic opportunity to leapfrog traditional lab models.

This report recommends a **hybrid, cloud-native, multi-vendor lab architecture** that combines:

- Container-native network operating systems (Containerlab + cEOS, cRPD, FRR, SONiC, SR Linux) for high-density, low-cost, GitOps-driven labs.
- Selective use of full virtual machines (via OpenShift Virtualization / KubeVirt) for Cisco CAT8000v, ISE, Palo Alto VM-Series, FortiGate, Juniper vSRX, etc., where full feature fidelity is required.
- EVE-NG / CML instances running as persistent VMs on OpenShift for CCIE-scale and complex multi-vendor topologies.
- Infrastructure-as-Code (YAML topologies, Ansible/Terraform, GitOps) so labs become version-controlled, reproducible, and CI/CD integrated.
- Open-source first approach to minimize licensing cost while still supporting official images where needed for certification accuracy.

Key Innovation: Treat the lab environment itself as a product — deliver “Lab-as-a-Service” on OpenShift with self-service namespaces, quota management, and automated teardown, borrowing the best patterns from Palo Alto (VM-Series + Panorama), Juniper (vMX/vSRX + vLabs), Fortinet (FortiGate VM + FortiManager), and modern open-source tools (Containerlab, freeRtr, Kathará).

2. Review of Provided Source Documents

2.1 Vendor Lab Environments Matrix Analysis

The matrix correctly prioritizes High-priority vendors (Cisco, Juniper, Fortinet, Check Point, Palo Alto, VMware, AWS, Google Cloud, IBM, Red Hat, Microsoft). It lists virtual platforms such as VIRT/CML, GNS3, IOSv, vSRX, vMX, FortiGate VM, VM-Series, Nested ESXi, OpenShift Cluster, etc. Implementation priority is sound, but the matrix still reflects a traditional “one VM per device” mindset. Missing or under-emphasized are:

- Container-native NOS options (cEOS, cRPD, FRR, SONiC, Nokia SR Linux) that scale far better on Kubernetes.
- Open-source pure software routers (freeRtr, FRRouting) for protocol labs without vendor image licensing issues.
- Lab orchestration patterns (Containerlab, Kathará, NetPilot-style AI assisted topologies).
- Integration of OpenShift Virtualization for mixed container + VM workloads.

2.2 CCIE EI Commercial Lab Offerings Analysis

The cicoland.net offerings (Dell R620 / T5610, 128–256 GB RAM, 1 TB SSD, ESXi 7.0, EVE-NG 6.2 + Viptela SD-WAN 20.12, CML-2 eval, ISE 3.1 demo, NSO, 10× CAT8000v) are excellent turnkey CCIE EI preparation platforms. Pricing (~\$1,000–\$1,150) is competitive for dedicated hardware. However:

- Hardware-centric model does not leverage your existing IBM Cloud / OpenShift investment.
- Licensing is evaluation / demo only for many components (CML free 5-node, ISE 90-day, Windows 180-day).
- Limited multi-vendor depth beyond Cisco + basic firewalls in the base image.
- No GitOps / automation-first design; labs are manually built and not easily shareable or versioned.
- Physical shipping and 4–5 day prep time introduce operational friction.

Recommendation: Use the commercial kits only for dedicated CCIE candidates who need offline/high-fidelity hardware. For the broader organization, migrate to a cloud-native OpenShift-based Lab-as-a-Service model that can host the same images (and many more) at lower ongoing cost and higher density.

3. Current Environment: IBM Cloud + Red Hat OpenShift Strengths

Running on IBM Cloud with Red Hat OpenShift gives you significant advantages over pure ESXi or bare-metal EVE-NG servers:

OpenShift Virtualization (KubeVirt): Run full VMs (CAT8000v, Palo Alto VM-Series, FortiGate, vSRX, Windows, ISE) side-by-side with containers in the same cluster.

User-Defined Networks (UDNs): Available in recent OpenShift versions; perfect for isolating lab topologies with deterministic IP addressing per namespace.

Multi-tenancy & Quotas: Give each engineer / team a namespace with resource quotas; auto-cleanup via operators or CronJobs.

GitOps (Argo CD / OpenShift GitOps): Treat every lab topology as code stored in Git; promote labs through environments.

Operators & Operators Framework: Build or reuse operators for lab lifecycle (start, stop, snapshot, share).

Observability: Native Prometheus, Grafana, Network Observability Operator for lab traffic analysis.

IBM Cloud Integration: Easy hybrid connectivity back to on-prem or other clouds for hybrid lab scenarios.

4. Recommended Architecture: Hybrid Multi-Vendor Lab on OpenShift

Adopt a three-tier lab model that balances fidelity, density, cost, and automation:

Tier 1 — Container-Native (Highest Density / Lowest Cost)

Use **Containerlab** (or Kathará) running inside OpenShift (or on dedicated worker nodes). Ideal for protocol labs, BGP, OSPF, EVPN, SONiC, FRR, Arista cEOS, Juniper cRPD, Nokia SR Linux. Labs defined in YAML, version-controlled, spun up in seconds. Perfect for CI/CD validation of network automation.

Tier 2 — Full Virtual Machines via OpenShift Virtualization

Deploy KubeVirt VMs for devices that require full OS images: Cisco CAT8000v / CSR1000v, Palo Alto VM-Series + Panorama, Fortinet FortiGate + FortiManager / FortiAnalyzer, Juniper vSRX / vMX, Check Point, F5 BIG-IP VE, Cisco ISE, Windows servers, etc. Connect them via Multus CNI or User-Defined Networks for multi-homed topologies.

Tier 3 — Persistent Shared Emulation Platforms

Run one or more long-lived EVE-NG Pro or CML instances as VMs (or bare-metal nodes) on the cluster. Use these for complex CCIE-scale topologies, SD-WAN (vManage/vBond/vSmart + CAT8000v), and multi-vendor labs that need the rich EVE-NG GUI and image library. Teams access via browser; images and topologies can still be managed via Git.

***Innovation Opportunity:** Build a thin “Lab Orchestrator” service (Python/Go operator or simple web UI) that accepts a high-level description or YAML, then deploys the correct mix of containers + VMs + EVE-NG nodes, wires them together, and publishes access URLs / credentials. This mirrors the direction of modern commercial platforms while remaining fully under your control.*

5. Comprehensive Catalog: Open-Source, Emulators & Virtual Platforms by Vendor

Below is an expanded, prioritised catalogue derived from your matrix, augmented with current best open-source and commercial virtual options (2025–2026 landscape).

Vendor / Category	Recommended Platforms	Open Source / Free Options	Notes / Priority
Cisco (Networking)	CML 2.x, CAT8000v, CSR1000v, IOSv, IOL, NX-OSv, ASAv, FTDv, ISE, NSO, Catalyst Center (DNAC) Simulator	GNS3 + Dynamips/IOU, Containerlab (IOL/XRd), freeRtr, Packet Tracer (limited)	High – Core of CCIE EI / Enterprise
Juniper	vSRX, vMX, vQFX, cRPD, vJunos-switch / router	Juniper vLabs (free hosted), Containerlab + cRPD, freeRtr	High – Excellent container story
Palo Alto	VM-Series, Panorama, Prisma Access lab tenants	Official trial images, Containerlab BYOI, community PAN-OS simulators	High – Security focus
Fortinet	FortiGate VM, FortiManager, FortiAnalyzer, FortiSandbox	Official evaluation images, Containerlab support	High – NSE path
Arista	vEOS, cEOS (container)	cEOS is free for lab use; Containerlab native	Medium-High – Ideal for containers
Check Point	CloudGuard IaaS / Security Gateway VMs	Evaluation images	High
F5	BIG-IP Virtual Edition	Trial licenses	Medium
VMware / Broadcom	Nested ESXi, VCSA, NSX-T	OpenShift Virtualization as alternative	High for virtualization labs
Red Hat / OpenShift	OpenShift itself, OpenShift Virtualization, RHEL VMs	OKD (upstream), freeRtr, FRRouting	High – Your platform
IBM	Power Systems LPARs (if available), IBM Cloud VPC labs, Watson / Cloud Pak demos	Open-source alternatives where possible	High – Leverage existing relationship
AWS / GCP / Azure	LocalStack, GCP Emulator, Azure simulators, official cloud sandboxes	LocalStack (open source), MinIO, etc.	High for multi-cloud
Linux / Open Source Routing	FRRouting, freeRtr, SONiC, Cumulus (NVIDIA), Linux bridges / VRF	All fully open source; Containerlab + Docker	Strategic – Cost & density winner
Monitoring / SIEM	Splunk Enterprise trial, SolarWinds, Prometheus + Grafana + Loki	Prometheus stack is fully open source	Medium
Collaboration / UC	CUCM (OVA), Webex, Zoom sandboxes, Avaya / Mitel VMs	Limited open source; use trials	Lower priority
Storage / Backup	NetApp ONTAP sim, Veeam, Rubrik, Pure //X Virtual	Trial images primarily	Medium
Endpoint / Cloud Security	CrowdStrike Falcon labs, Zscaler demo tenants, Kaspersky, Sophos XG	Vendor demo tenants preferred	Medium

Best overall open-source / free stack for 2026: Containerlab (primary orchestrator) + FRRouting / freeRtr / cEOS / cRPD / SONiC + GNS3 or EVE-NG Community for classic images + OpenShift Virtualization for full VMs.

6. Best Ways to Obtain Open-Source Tools & Emulators

6.1 Fully Open-Source / Free Platforms

- **Containerlab** — <https://containerlab.dev> — Best modern choice. YAML topology, Docker-native, excellent documentation, supports 20+ NOS images.
- **GNS3** — <https://www.gns3.com> — Mature GUI, huge community image library, still excellent for desktop / small team use.
- **EVE-NG Community** — Free (node limits apply in newer versions); Pro is inexpensive for team use.
- **freeRtr** — <https://www.freertr.org> / GitHub mc36/freeRtr — Pure software multi-protocol router, extremely flexible.
- **FRRouting (FRR)** — Industry standard open-source routing stack; runs natively in containers.
- **Kathará** — Lightweight Docker-based network emulation, good for teaching.
- **SONiC** — Microsoft's open networking switch OS; runs in containers.
- **ns-3 / CORE / Cloonix** — More academic / research oriented simulators.

6.2 Official / Evaluation Images (Legitimate Paths)

- Cisco: CML Personal / Personal Plus (includes official images), DevNet sandboxes, Cisco Modeling Labs free 5-node tier, evaluation images via Cisco account.
- Juniper: Juniper vLabs (free hosted), evaluation images for vSRX / vMX via Juniper account.
- Palo Alto: VM-Series evaluation licenses + Panorama trials.
- Fortinet: FortiGate VM evaluation images (time-limited).
- Arista: cEOS is freely available for lab use.
- Always prefer official evaluation or licensed images for certification accuracy and legal compliance. Avoid grey-market IOS images.

6.3 Hosted / Cloud Lab Services (Quick Start)

- Vendor free sandboxes: Cisco DevNet, Juniper vLabs, Palo Alto / Fortinet demo tenants, Zscaler, etc.
- Commercial Lab-as-a-Service: CloudMyLab (hosted EVE-NG), emerging AI-assisted platforms (e.g. NetPilot-style).
- Your own OpenShift: Best long-term option for control, cost, and integration with IBM Cloud.

7. Implementation Roadmap (Phased)

Phase 0 – Foundation (2–4 weeks)

- Enable OpenShift Virtualization (KubeVirt) on the IBM Cloud OpenShift cluster.
- Create dedicated lab projects/namespaces with resource quotas and network policies.
- Deploy Containerlab on a set of worker nodes or as a privileged DaemonSet / Job pattern.
- Set up Git repository structure for topologies (Containerlab YAML + Ansible playbooks).

Phase 1 – Quick Wins (4–6 weeks)

- Stand up Tier-1 container labs: FRR, freeRtr, Arista cEOS, Juniper cRPD, SONiC.
- Import 2–3 high-priority full VMs (CAT8000v, FortiGate, Palo Alto VM-Series) via OpenShift Virtualization.
- Document access patterns (console, SSH, web UI) and create simple self-service templates.
- Pilot with 5–10 engineers on common CCNA/CCNP / NSE / PCNSA scenarios.

Phase 2 – Shared Platforms (6–10 weeks)

- Deploy one or two EVE-NG Pro (or CML) instances as persistent VMs.
- Migrate key CCIE EI topologies (including SD-WAN controllers) into the shared platform.
- Integrate with OpenShift GitOps for topology promotion and cleanup.
- Add observability (Prometheus exporters on lab devices where possible).

Phase 3 – Lab-as-a-Service (Ongoing)

- Build a lightweight portal or operator that lets users request “I need a 6-node BGP + 2 firewall lab” and receives a ready topology.
- Implement auto-teardown (TTL) and cost reporting per namespace.
- Expand image library systematically following the priority order in the vendor matrix.
- Explore hybrid connectivity so labs can reach real cloud resources or on-prem hardware when needed.

8. Strategic Innovations & Differentiators

- **GitOps-first labs:** Every topology is a Git commit. Peer review of lab designs becomes possible; regressions are caught in CI.
- **Mixed fidelity:** Use containers for the “boring” protocol work and full VMs only where feature depth is required — dramatic density gains.
- **Certification-aligned image sets:** Maintain curated image bundles that match exact CCIE EI, NSE, PCNSA, JNCIA versions.
- **Lab cost attribution:** Use OpenShift cost management / IBM Cloud billing tags so training budgets can be tracked.
- **AI-assisted topology generation (future):** Explore natural-language → Containerlab YAML pipelines (emerging 2026 capability).
- **Cross-vendor integration labs:** Intentionally design scenarios that force Cisco + Palo Alto + Fortinet + Juniper interoperability — these are rare and highly valuable for real-world readiness.
- **Security of the lab itself:** Treat lab namespaces as untrusted; use NetworkPolicies, restricted SCCs, and image scanning so a compromised lab image cannot escape.

9. Cost, Licensing & Risk Considerations

Commercial Dell-based CCIE kits are attractive for individuals but do not scale for an organization. On OpenShift you pay primarily for compute (vCPU/RAM) and storage. Container-native devices consume far less memory than full QEMU VMs. Evaluation licenses remain time-limited; plan for permanent licenses only for the most critical platforms (or rely on official free tiers such as CML 5-node, Arista cEOS, Juniper vLabs).

Legal note: Always obtain vendor images through legitimate channels (customer portal, evaluation requests, DevNet, etc.). Do not distribute or store unauthorized IOS / PAN-OS / FortiOS images.

10. Conclusion & Immediate Next Steps

You already possess the two most important ingredients: (1) a modern Kubernetes platform (OpenShift on IBM Cloud) and (2) a clear prioritised list of vendors and certifications. By shifting from a pure ESXi + EVE-NG hardware model to a hybrid container + KubeVirt + selective EVE-NG model, you can deliver higher density, lower cost, better automation, and true multi-vendor coverage while still supporting the high-fidelity CCIE EI requirements.

Immediate recommended actions:

- Approve a small pilot project (2–3 engineers, 4 weeks) to stand up Containerlab + 2–3 KubeVirt lab VMs on the existing OpenShift cluster.
- Inventory current legitimate image access (Cisco, Palo Alto, Fortinet, Juniper accounts) and open evaluation requests where gaps exist.
- Create a Git repository skeleton for lab topologies and a simple naming / tagging convention.
- Schedule a follow-up architecture workshop to finalise namespace strategy, resource quotas, and the Lab-as-a-Service roadmap.
- Optionally evaluate one commercial hosted EVE-NG service as a short-term bridge while the OpenShift platform matures.

This approach positions i3 Smart Labs as a modern, cloud-native training and validation platform rather than a collection of isolated physical servers. It directly supports the High-priority vendors in your matrix while dramatically expanding what is possible with open-source and evaluation software.

*Prepared by Senior Network Infrastructure Consulting practice.
For questions or implementation support, contact the architecture team.*